
Measurement Pitfalls in Evaluating Learned Hedging Policies

Aditya Bhatia

Stevens Institute of Technology

[AUTHOR -- email]

Abstract

Deep hedging trains a neural policy to minimise a convex risk measure of simulated profit and loss, so its evidence is statistical: a learned policy is judged better when a tail risk measure is lower, across training seeds, in a simulated market. We measure five ways that evidence misleads while passing the obvious check, in an open pipeline where every number is regenerated by committed code. (1) In TensorFlow 2.21, consecutive integer seeds passed to `from_seed` yield one Philox stream shifted by four draws per seed, bypassing a guard in the library’s own source; replicate dispersion collapses to $0.41\times$ the sampling error, and a $+0.85$ standard-error deviation reads as $+9.3$. (2) Trading a Whalley–Wilmott band back to its centre forfeits 0.099 of CVaR-95 improvement ($t = 10.9$, 20/20 paired seeds). (3) Across 12 contracts, a 10% adverse volatility error moves CVaR-95 $2.7\text{--}7.1\times$ as much as a 5 bp cost. Mean effects match closed forms to 3.5%, but the tail amplifies the two frictions differently, so at 25 bp the dominant friction depends on the risk measure; in market data such an error occurs about one month in eleven, while hedging a liquid underlying costs under 2 bp. (4) From a 30-replicate training-seed noise floor, five seeds detect an effect the size of band-versus-delta with probability 0.36, and 80% power needs 12 seeds against a fixed baseline or 20 per arm between learned policies. Quadrupling the training budget more than halves the typical spread, but one run in 30 destabilised and alone more than doubled the s.d., passing the usual convergence check. (5) One trained network passes or fails a pre-specified accuracy gate depending on two unstated evaluation choices. For each pitfall we give an executable check, and we record the errors in our own analyses that these checks caught.

1 Introduction

Deep hedging [Buehler et al., 2019] replaces the hedge implied by a pricing model with a neural policy trained to minimise a convex risk measure of simulated terminal profit and loss (P&L). Whether a learned policy beats a classical rule is then an empirical question answered by simulation: tail risk estimated on sampled paths, compared across training seeds, under a chosen market model and friction. Every link in that chain — the random streams that make replicates independent, the classical baseline, the frictions a benchmark varies, the number of replicates, and the acceptance tests that decide when a policy has been learned — can fail in a way that no single run reveals. Each failure changes a conclusion, not a decimal place.

The failure classes are known in general. Analysing non-independent replicates as independent is pseudoreplication [Hurlbert, 1984]. Counter-based generators give independent streams only through distinct keys or disjoint counters [Salmon et al., 2011], and TensorFlow’s guide warns that seeding generators with `from_seed` risks overlapping streams [TensorFlow Authors, 2024]. Few seeds mislead in deep reinforcement learning [Henderson et al., 2018, Colas et al., 2018, Agarwal

et al., 2021], and correlated replicates inflate the variance of benchmark estimates [Bouthillier et al., 2021]. Neural-hedging evaluations have been questioned for their choice of benchmark and data splits [Ruf and Wang, 2020], for lacking a standard benchmark [Pickard and Lawryshyn, 2023], and for comparing against lightly cost-adjusted classical rules [Kumar, 2026]. What is missing is measurement: how large each failure is where learned hedgers are evaluated, and which check catches it. Deep hedging is a good laboratory for this, because it has closed-form references — the Black–Scholes delta, the Whalley–Wilmott band [Whalley and Wilmott, 1997], Leland’s cost adjustment [Leland, 1985] — and unlimited replicates.

We measure five such pitfalls. Each is a plausible evaluation that passes the obvious check; for each we report its magnitude, the check that catches it, and a fix. Our contribution is quantification and evaluation design, not the discovery of new failure classes; §9 places each against prior work.

Contributions. Each is a measurement in our own pipeline, not a claim about any published study; Appendix A maps every number to the command that regenerates it.

1. **Pseudo-replicates (§3).** TensorFlow 2.21’s integer-seed path writes the seed into the Philox counter under a zero key, so consecutive seeds give shifted copies of one stream; a guard in the library’s source that exists to prevent exactly this never fires for an integer. The consequence is false significance, not bias (+0.85 standard errors reported as +9.34). A list seed or a hashed seed avoids it; a dispersion test detects it where a lag-0 correlation cannot.
2. **A band that isn’t a band (§4).** The paired 20-seed CVaR-95 cost of implementing the Whalley–Wilmott band with a trade back to its centre (0.099, $t = 10.9$), and a demonstration that the natural summary ratio is not estimable.
3. **The wrong friction (§5).** A 12-contract comparison of volatility error with proportional cost, decomposed against closed forms: which dominates depends on the direction of the error, the cost level, the rebalancing frequency and the risk measure. Both axes are calibrated to 36 years of S&P 500 data and to quoted spreads.
4. **Too few seeds (§6).** A 30-replicate training-seed noise floor at two training budgets, power-based seed counts (12 against a fixed baseline, 20 per arm between learned policies), and a failed run that both a mean $\pm$ s.d. summary and the usual convergence check miss.
5. **An underspecified acceptance gate (§7).** The verdict of a pre-specified accuracy test flips with two unstated evaluation choices, because trained policies depend on an input the optimal policy ignores; an invariance check makes the gate well posed.

The checks we propose are properties of the *evidence behind a comparison* rather than of any single output, and they apply whoever or whatever wrote the code. Four of the five pitfalls first appeared in this project’s own pipeline or drafts and were caught by the checks we describe; we record each correction where it occurred.

2 Setup

P&L. An agent is short one European call with payoff $Z = (S_n - K)^+$ and receives a premium p_0 at $t_0 = 0$. It holds δ_i shares over $[t_i, t_{i+1})$, $t_i = iT/n$, chosen with information available at t_i . With proportional cost rate c ,

$$\text{PL} = p_0 - Z + \sum_{i=0}^{n-1} \delta_i (S_{i+1} - S_i) - c \sum_{i=0}^n S_i |\delta_i - \delta_{i-1}|, \quad \delta_{-1} = \delta_n = 0. \quad (1)$$

The cost sum runs to n : the position is opened at t_0 and liquidated at T , so n decisions make $n + 1$ trades; without the terminal term a policy could carry any position into expiry at no charge. The benchmark is specified in discounted units, under which (1) is form-invariant; all experiments here use $r = 0$. Equation (1) is implemented once (dhbench/pnl.py) and every P&L in this paper is computed by that function.

Risk measures. We report CVaR-95, the mean of the worst 5% of outcomes, by sorting and averaging. A CVaR training objective uses the Rockafellar–Uryasev form $\min_w \{w + (1 - \alpha)^{-1} \mathbb{E}[(-X -$

Table 1: The mechanism in TensorFlow 2.21.

Quantity	Value
State after <code>from_seed(k)</code>	$[k, 0, 0]$
State after <code>from_seed([k])</code>	$[0, 0, k]$
Seed- k stream vs. seed 0	offset $4k$
Seed-1 draws in seed-0 stream	100%
Lag-0 corr., seeds 0 and 1	+0.0051

Table 2: ATM call by Monte Carlo: 20 replicates of 200,000 paths, 50 steps (truth 7.96557).

	<code>from_seed(k)</code>	Hashed
Mean of the 20 replicates	7.99070	7.95994
Cross-replicate s.d.	0.01203	0.03429
Analytic SE of one estimate	0.02953	0.02943
Dispersion / analytic SE	0.41	1.16
Deviation, SEs of one estimate	+0.85	-0.19
Deviation, naive SEs of the mean	+9.34	-0.73
Replicates above the truth	20/20	7/20
Corr. of log S_T , replicates 0, 1	0.920	0.004

$w)^+]$ [Rockafellar and Uryasev, 2000] with w trained jointly; the two estimators agree at the optimum (20.6004 on a 200,000-sample Gaussian test case). Trained policies minimise the entropic risk $\lambda^{-1} \log \mathbb{E}[e^{-\lambda \text{PL}}]$. Tables state their orientation: *P&L* (higher is better; harmful effects are negative) or *loss* (lower is better).

World and baselines. Prices follow geometric Brownian motion, simulated exactly in log space with all normal draws of a run taken in one call of shape (paths, steps). Unless stated, $S_0 = K = 100$, $T = 1$, $\sigma = 0.2$, $\mu = r = 0$, and the premium is the Black–Scholes price. The baselines are the Black–Scholes delta $\Phi(d_1)$, rebalanced at every date, and the Whalley–Wilmott band of half-width $H = (\frac{3}{2} c S \Gamma^2 e^{-r\tau} / \lambda)^{1/3}$ around $\Phi(d_1)$, with $\lambda = 1$, implemented as a clip, $\delta_i = \text{clip}(\delta_{i-1}, \Phi(d_1) - H, \Phi(d_1) + H)$, so that no other rebalancing target is representable in the baseline code (§4).

Verification ladder. Results are interpreted only after the layers beneath them pass executable checks. The Monte Carlo call price matches its closed form within three standard errors (SEs). The zero-cost delta hedge sold at the Black–Scholes price gives $\text{std}(\text{PL})\sqrt{n} = 6.64, 6.87, 6.92, 7.02$ at $n = 10, 40, 160, 640$, the $n^{-1/2}$ law, which holds only if simulator, pricing reference and accounting agree. Under 50 bp costs the band beats every-step delta hedging on both CVaR-95 and turnover. A further check arrived with Pitfall 3: across 12 contracts the simulated *mean* effects of a volatility error and of proportional cost match closed forms within 0.3% and 3.5% (§5). Pitfalls 1–3 involve no neural network.

3 Pitfall 1: pseudo-replicates

Seed-averaged error bars assume that replicate seeds give independent random streams. Seeding replicate r with `tf.random.Generator.from_seed(r)` violates this in TensorFlow 2.21.

Mechanism. Philox [Salmon et al., 2011], TensorFlow’s default generator, is counter-based: its state holds a counter and a key, distinct keys give distinct streams, and each counter value yields four outputs. `from_seed(k)` with a Python integer sets the state to $[k, 0, 0]$ — k in the counter, the key zero for every seed — so the stream of seed k is the stream of seed 0 with its first $4k$ draws removed (Table 1). TensorFlow documents the seed-to-state conversion as unspecified, and its guide warns that constructing generators with `from_seed` risks “seeds that lead to overlapping random-number streams” [TensorFlow Authors, 2024]. Its source is more specific: seeds are padded with zeros on the *left* because padding on the right “would cause a small seed to be used as the ‘counter’ while the ‘key’ is always zero”, in which case “two RNGs with two different small seeds may generate overlapping outputs” [TensorFlow Authors, 2026]. The guard works for a list seed — `from_seed([k])` gives $[0, 0, k]$, a distinct key — but an integer is first split into as many words as the state has, arrives at full length, and is never padded. The most natural call therefore realises, deterministically and for every pair of small seeds, the overlap the comment was written to prevent. Related seeds yielding correlated streams is a long-known defect class [Matsumoto et al., 2007]. This instance passes routine checks: each replicate reproduces bit for bit, and the lag-0 correlation of the seed-0 and seed-1 streams is +0.0051, because shifting an i.i.d. sequence decorrelates it pointwise while leaving the set of numbers unchanged.

Consequence: false significance, not bias. Price an at-the-money call (closed form 7.96557) with 200,000 paths of 50 steps under `from_seed` seeds $0, \dots, 19$. Path p of replicate $k + 1$ reuses 46 of the 50 increments of path p of replicate k , and the 20 replicates together use only 76 more distinct normal draws than one. The replicate mean deviates from the truth by $+0.85$ SEs of a single estimate: ordinary luck, on the right yardstick for what is essentially one sample. Treated as 20 independent replicates, the same deviation is divided by their standard error of the mean, $0.41 \times 0.02953/\sqrt{20} = 0.00269$, and reads as $0.85\sqrt{20}/0.41 = 9.34$ SEs, with all 20 replicates above the truth (Table 2). The dispersion ratio also measures the dependence: for replicates with average pairwise correlation $\bar{\rho}$ the expected sample variance is $(1 - \bar{\rho}) \sigma^2$, so a ratio of 0.41 implies $\bar{\rho} \approx 0.83$. This is the correlated-replicate inflation analysed by Bouthillier et al. [2021, Eq. 7], arising here silently from the seeding path rather than from a deliberate choice; in Hurlbert’s terms, the replicates exist but are not independent, and the error term is wrong.

A correction we record deliberately. An earlier write-up in this project called the $+0.025$ a *bias* propagated through a “systematic offset” in realised volatility. The numbers were right and the mechanism was wrong: nothing is biased; 20 pseudo-replicates of one sample were counted as 20 samples. The bias reading fitted every summary statistic we had computed; only reading the generator state directly distinguished the two.

Fix and check. We map replicate indices to seeds by SHA-256 of a string holding a stream label (“train”, “eval”, “init”) and the index, truncated to 63 bits; the labels make training and evaluation randomness disjoint by construction, and SHA-256 replaces Python’s `hash()`, which is salted per process. Hashed seeds still land in the counter under key zero; the fix works because they scatter counters across $[0, 2^{63})$. Passing a list, `from_seed([k])`, puts k in the key instead and is the one-character alternative; TensorFlow recommends `Generator.split` for independent streams [TensorFlow Authors, 2024]. With hashed seeds the experiment gives dispersion 1.16 and a deviation of -0.19 SEs. The check that catches the defect is the dispersion of a downstream estimator against its analytic SE: a test requires $[0.6, 1.7] \times$ across 16 replicates of 60,000 paths (unhashed seeding scores 0.34 and fails). Further tests pin the counter layout and the list-seed key, so a change in TensorFlow’s seeding fails loudly.

4 Pitfall 2: a band that isn’t a band

Beating every-step delta hedging under costs requires only trading less, so the informative classical comparator for a learned policy is a cost-aware band. The asymptotically optimal rule under proportional costs holds the position while it lies inside a band around $\Phi(d_1)$ and, on exit, trades “so as to just stay inside” [Whalley and Wilmott, 1997, working paper, p. 6] — to the nearest edge; an interior rebalance point is optimal only when costs have a fixed component [Whalley and Wilmott, 1999]. Rehedging back to delta on a breach is a different strategy: Whalley and Wilmott’s “market movement” model, “a strategy commonly used in practice” (working paper, p. 18), which remains a standard rule-based hedger in recent learning work [Chen et al., 2025]. The pitfall is to implement the second and call it the first. It is easy to make: two of this project’s own design documents specified it until a review caught it, before the baseline was implemented. The implementations we inspected trade to the edge [Imaki et al., 2023, Kumar, 2026, Sakuma, 2026], and Sakuma [2026, App. E] shows that trigger-to-centre execution raises turnover and reserve cost. We measure what the choice does to the baseline a learned policy is judged against, on the tail metric deep hedging optimises.

Table 3 compares the three strategies on the ATM one-year call ($n = 50$, $c = 50$ bp, $\lambda = 1$) over 20 seeds of 20,000 paths, with paths and premium shared within each seed. The centre-rule baseline sits closer to delta hedging (0.035 away) than to the correct band (0.099 away), so a learned policy L measured against it gains 0.099 of apparent advantage, in expectation, from the baseline alone: $L - C = (L - E) + (E - C)$.

Why we report no ratio. Table 3 invites a summary as the fraction of the band’s improvement that the centre rule captures. That ratio is not estimable here: its numerator is comparable to its own seed-to-seed spread (mean 0.0354, s.d. 0.0327), and across the 20 seeds the ratio has mean 23.7%, s.d. 26.5% and range -35.9% to $+67.9\%$. An earlier single-seed run of this project, with unhashed seeds, quoted “7%”. We withdrew it, and a test now asserts that the ratio remains unstable so that it

Table 3: Paired CVaR-95 differences (P&L orientation; positive favours the first strategy). ATM one-year call, 50 bp cost, 20 seeds of 20,000 paths.

Comparison	Mean	S.d.	t	Same sign
Edge band – delta hedge	+0.1343	0.0446	13.46	20/20
Centre band – delta hedge	+0.0354	0.0327	4.84	17/20
Edge band – centre band	+0.0989	0.0405	10.93	20/20

Table 4: Effect on CVaR-95 (P&L orientation; negative is worse) against the correctly specified zero-cost delta hedge; mean of 3 paired seeds of 40,000 paths. Volatility columns: $\sigma_r = \rho \sigma_h$, zero cost. Cost columns: $\sigma_r = \sigma_h$. Ratio: absolute effect at $\rho = 1.1$ over absolute effect at 5 bp. W/D: weekly/daily rebalancing, n steps. S.d.s in Appendix B.

K	T		n	Volatility error ρ			Proportional cost				Ratio
				0.9	1.1	1.25	1 bp	5 bp	25 bp	50 bp	
90	0.25	W	13	+0.634	-0.687	-1.792	-0.026	-0.131	-0.664	-1.347	5.2
90	1	W	52	+1.086	-1.232	-3.241	-0.041	-0.206	-1.052	-2.142	6.0
100	0.25	W	13	+0.743	-0.799	-2.073	-0.027	-0.133	-0.679	-1.386	6.0
100	1	W	52	+1.222	-1.370	-3.598	-0.043	-0.218	-1.114	-2.274	6.3
110	0.25	W	13	+0.768	-0.829	-2.164	-0.023	-0.117	-0.598	-1.227	7.1
110	1	W	52	+1.296	-1.471	-3.872	-0.044	-0.221	-1.134	-2.320	6.7
90	0.25	D	63	+0.466	-0.559	-1.496	-0.042	-0.210	-1.087	-2.234	2.7
90	1	D	252	+0.849	-1.136	-3.032	-0.075	-0.382	-2.003	-4.121	3.0
100	0.25	D	63	+0.601	-0.681	-1.790	-0.046	-0.235	-1.218	-2.503	2.9
100	1	D	252	+0.984	-1.276	-3.397	-0.082	-0.416	-2.185	-4.498	3.1
110	0.25	D	63	+0.572	-0.681	-1.818	-0.041	-0.210	-1.110	-2.302	3.2
110	1	D	252	+0.989	-1.359	-3.637	-0.084	-0.429	-2.265	-4.676	3.2

cannot return unnoticed. Differences in levels, with their paired dispersion and sign consistency, are the defensible statement.

5 Pitfall 3: the wrong friction

A benchmark that holds volatility known and correct while sweeping transaction costs presumes that costs are the first-order friction. Classical theory says which way each friction pushes. Leland’s adjustment prices a proportional cost as extra variance [Leland, 1985]; a convex claim hedged at too high a volatility is super-replicated [El Karoui et al., 1998], because the hedged position accrues P&L at rate $\frac{1}{2}\Gamma S^2(\sigma_h^2 - \sigma_r^2)$ with $\Gamma \geq 0$. Theory does not say how the two compare in the tail metric a deep-hedging benchmark reports. We measure that for the classical delta hedge; the result concerns hedging economics rather than any learning method, and it decides which axes a benchmark must vary.

Design. Twelve contracts: strikes 90, 100, 110 ($S_0 = 100$), maturities 0.25 and 1 year, weekly or daily rebalancing. The hedger computes deltas and charges the premium at $\sigma_h = 0.2$ (charging it at the realised volatility would hide the misspecification inside the premium). The market realises $\sigma_r = \rho \sigma_h$, $\rho \in \{0.8, 0.9, 1.1, 1.25\}$, at zero cost, or $\sigma_r = \sigma_h$ with proportional costs of 1, 5, 25 or 50 bp. An *effect* is the change in CVaR-95 (P&L orientation) against the correctly specified zero-cost hedge of the same contract. Within a contract and seed every variant is driven by the same normal draws, so effects are paired; with 3 seeds of 40,000 paths, every one of the 96 effects has $|\text{mean}|/(\text{s.d.}/\sqrt{3}) \geq 91$ (Appendix B).

Findings. *Adverse volatility error dominates small costs:* realised volatility 10% above the hedging volatility hurts CVaR-95 more than a 5 bp cost in all 12 contracts, by 2.7 to 7.1 $\times$. *The effect is directional:* realised volatility 10% below it is a gain in all 12 (+0.47 to +1.30), as the super-replication argument predicts, and in every contract the loss at $\rho = 1.1$ exceeds the gain at $\rho = 0.9$, so averaging a symmetric perturbation understates the adverse case. *The ordering depends on the cost regime:* at 25 bp it splits exactly by rebalancing frequency (the +10% volatility effect is larger

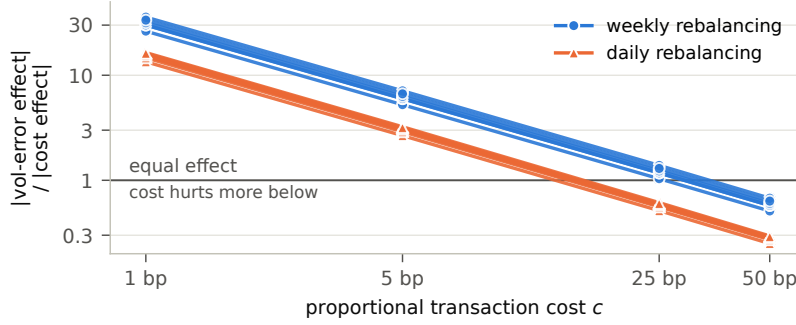


Figure 1: Ratio of the CVaR-95 effect of a +10% volatility error to that of proportional cost c , for the 12 contracts (lines join the four measured cost levels); above 1 the volatility error hurts more. Weekly and daily contracts form two bundles that cross 1 between 25 and 50 bp and between 5 and 25 bp, respectively. Drawn by `paper/workshop/make_figure.py` from `paper/findings_sweep.json`.

in all six weekly contracts, the cost effect in all six daily ones); at 50 bp the cost effect exceeds the +10% volatility effect in all 12. Moving from weekly to daily rebalancing multiplies the 5 bp cost effect by 1.6–1.9 while the volatility effect shrinks by 7–19%, so the ratio falls from $5.2\text{--}7.1\times$ to $2.7\text{--}3.2\times$; the pooled median, $4.2\times$, lies between the two clusters and describes no contract (Figure 1). Unlike the ratio of §4, these ratios are well defined: both effects sit far from zero relative to their noise.

Mechanism: the ordering belongs to the tail. On the *mean*, closed forms hold in all 12 contracts. With $\mu = r = 0$ the mean effect of the volatility error is exactly $V(\sigma_h) - V(\sigma_r)$, matched to 0.3%; the mean cost is Leland’s rebalancing cost plus the cost of opening and closing the hedge, matched to 3.5% (Appendix C). This is an analytic check of the simulator and of the accounting, including the opening trade and the terminal liquidation, which together carry 13–75% of the cost. But CVaR-95 is not a rescaled mean. It amplifies the volatility effect $1.6\text{--}3.0\times$ relative to its mean and the cost effect only $1.1\text{--}2.1\times$, so the ordering in the tail differs from the ordering on the mean: at 25 bp the cost effect is the larger in 10 of 12 contracts on the mean, but only in the 6 daily contracts on CVaR-95. Which friction a benchmark finds first-order is a property of its risk measure as well as of the market. A shortcut makes the point sharper. Leland’s rebalancing-only ratio, $((1.1)^2 - 1)/\text{Le}$ with $\text{Le} = c\sqrt{8/\pi}/(\sigma\sqrt{\Delta t})$, tracks the CVaR-95 ratios to within 28% — but only because the two things it omits, the opening and closing trades and the tail amplification, roughly offset. An analytic prediction that agrees with a simulation can agree for the wrong reason; decomposing the agreement is the check.

Calibration. Both axes were set by assertion; we calibrate them from public data (Appendix D). For the volatility error we compare realised S&P 500 volatility over the 21 trading days after each close with VIX at that close, 1990–2026 (9,227 windows). The median ratio is 0.73, and realised volatility is at least 10% *below* VIX in 78% of windows, the favourable direction. It exceeds VIX by 10% or more in 8.8% of windows (block-bootstrap 95% CI 6.3–11.4%) and by 25% in 4.9%: a +10% adverse error is about the 91st percentile, one month in eleven, and adverse months cluster at shock onsets. For costs, a one-tick half-spread is 0.16 bp for E-mini S&P 500 futures and 0.065 bp for SPY at current index levels, and 1.6 bp for the median S&P 500 constituent in 2015; 25–50 bp is small- and micro-cap territory. VIX is not the volatility at which a desk hedges a given strike, and one-year contracts are not calibrated here, so these are anchors rather than a joint model.

Implication. For liquid underlyings, the adverse volatility error of a one-month-in-eleven event costs more tail risk than any realistic proportional cost — at the E-mini’s quoted spread the sweep’s 5 bp corresponds to a quote 31 ticks wide — so a protocol that fixes volatility while sweeping costs varies the second-order axis. Costs dominate only for illiquid underlyings under frequent rebalancing. Because adverse volatility errors concentrate in market stress, when spreads tend to widen, the two axes are not independent either. A benchmark should vary both, report its conclusion under the risk measure it claims, and separate adverse from favourable errors.

Table 5: Resolution and power implied by the training-seed noise floor at 2,000 steps ($s = 0.144$, 30 replicates) for an effect the size of band-versus-delta (0.134, Table 3). Power is for a two-sided t -test at $\alpha = 0.05$, from the noncentral t .

Seeds k	$t_{0.975, k-1}$	95% CI	Power	
		half-width	vs. fixed baseline	learned vs. learned
5	2.776	0.179	0.36	0.26
6	2.571	0.151	0.46	0.31
10	2.262	0.103	0.75	0.51
12	2.201	0.091	0.84	0.59
20	2.093	0.067	0.98	0.82

6 Pitfall 4: too few seeds

How many replicates a comparison needs depends on which source of variability dominates, and on the power it demands.

Evaluation noise. Over 100 independent replicates of the zero-cost delta hedge, the SE of CVaR-95 is $3.9\text{--}4.3\times$ that of the mean at equal path count (0.0569, 0.0268 and 0.0131 at $N = 5,000$, 20,000 and 100,000 paths, against 0.0138, 0.0069 and 0.0031), so path counts must be set by the tail metric. Pairing strategies on common paths helps the mean more than the tail. For band against delta at 50 bp (12 seeds of 20,000 paths), pairing shrinks the s.d. of the difference $1.42\times$ for mean P&L but only $1.19\times$ for CVaR-95 (paired s.d. 0.0480), because the two strategies’ tails are populated by different paths. Only when one strategy is compared with itself under a perturbed parameter (§5) does pairing make effects nearly deterministic. A ratio of s.d.s estimated from few seeds is itself noisy: at 8 seeds of 10,000 paths the same two factors read 0.88 and 0.96.

Training noise. We trained 30 replicates of a feedforward policy (two hidden layers of 32 ReLU units; inputs τ/T , S/K and the previous position; unconstrained output) under the entropic objective ($\lambda = 1$) at zero cost, for 2,000 Adam steps on fresh batches of 512 paths. Each replicate had its own initialisation and training paths, and all were scored on the same 20,000 evaluation paths, isolating training variability. CVaR-95 (loss orientation) had mean 2.386, range [2.198, 2.665] and s.d. $s = 0.144$, 95% CI [0.115, 0.193]: three times the paired evaluation s.d. of the band-versus-delta comparison, so intervals from resampling test paths alone would understate the uncertainty of a learned-policy comparison about threefold.

Power. Table 5 gives, for k seeds, the half-width of a 95% CI and the power to detect an effect the size of band-versus-delta, both against a fixed comparator (one-sample t on k paired differences) and between two learned policies (two-sample t , k per arm). Five seeds detect it with probability 0.36. Power of 80% needs 12 seeds against a fixed comparator — 8 to 19 across the CI of s — and 20 per arm between learned policies; for half that effect, 39 and 74. A learned-versus-band difference may well be smaller than band-versus-delta, so these counts are floors. Colas et al. [2018] make the same point for deep reinforcement learning; what is new here are the domain quantities, for a tail-risk objective whose training-seed noise exceeds its evaluation noise.

Longer training, and a run that failed. At 8,000 steps the same 30 replicates improve (mean CVaR-95 2.193 against 2.386) and their typical spread shrinks by more than half: the robust s.d. (interquartile range/1.349) falls from 0.160 to 0.063. The s.d. barely moves, 0.134 (95% CI [0.107, 0.180]), because one replicate did not converge. Its training loss fell to about 0.65 by step 3,000, like the others, then rose to 1.4 by step 4,500 and settled near 1.0, while a typical run went on to 0.58; its CVaR-95 ended at 2.833, 11 robust s.d.s from the median, having been unremarkable at 2,000 steps (2.483, ninth worst of 30). Without it the s.d. is 0.059. A replicate’s score at 2,000 steps barely predicts its score at 8,000 (correlation 0.17), so early ranking would not have caught it. Nor did the convergence check this project first used — last-decile loss below first-decile loss, which the initialisation spike satisfies — whereas the ratio of final-window to best-window loss flags it (+55%, against about zero for converged runs). One failure in 30 is a rate with a wide interval (Clopper–Pearson 95%: 0.1–17%), but at the observed rate a 12-seed study contains a failed run

Table 6: One gate, three evaluation designs: deviation from $\Phi(d_1)$ over R , mean $\pm$ s.d. across 6 training replicates. Gate: MAD < 0.05 and max < 0.15 .

Evaluation design	MAD	Max deviation	Replicates passing
Grid, previous position = 0	0.053 ± 0.014	0.173 ± 0.038	1/6
Grid, previous position = $\Phi(d_1)$	0.034 ± 0.007	0.097 ± 0.011	6/6
States visited by the policy	0.027 ± 0.005	0.119 ± 0.017	6/6

with probability 0.33, and a comparison with 20 runs per arm with probability 0.74. A power calculation on the s.d. treats such a run as Gaussian noise, and it is not. Seed counts must be set at the training budget actually used, and failed runs counted by a criterion fixed before seeing outcomes, reported, and never dropped, with robust aggregates such as the interquartile mean [Agarwal et al., 2021] alongside the mean.

Two corrections we record. An earlier version of our analysis computed the seed requirement with a normal critical value of 2.0 and concluded that four seeds sufficed; $t_{0.975,4} = 2.776$ is 39% larger. A later version reported the 95% CI half-width as a “minimum detectable effect” and concluded that six seeds sufficed, but an effect equal to that half-width is detected only about half the time; and its s came from 8 replicates (0.122), whose own CI spanned 6 to 29 seeds. Tests now pin the t critical value, the power calculation (checked against 200,000 simulated t -tests per seed count), and the gap between a CI half-width and an 80%-power effect.

7 Pitfall 5: an underspecified acceptance gate

A verification ladder is only as good as its gates. Ours required, before any learned-policy result could be reported, that a policy trained at zero cost recover the Black–Scholes delta: mean absolute deviation (MAD) below 0.05 and maximum deviation below 0.15 against $\Phi(d_1)$ over the region $R = \{S/K \in [0.8, 1.2], \tau \in [0.1, 1]\}$. The gate was written before any policy was trained, and it looked complete. It is not. It leaves two choices unstated: (i) the value of the previous-position input when the policy is evaluated on a grid of (τ, S) — the optimal zero-cost hedge ignores that input, but the network receives it — and (ii) how the states in R are weighted, uniformly on a grid or as the policy visits them.

Table 6 scores six replicates (8,000 Adam steps each, otherwise as in §6) under three designs over the same region: a grid with the previous position set to 0; a grid with it set to $\Phi(d_1)$, close to the position a well-trained policy holds along its own paths; and the states visited by the policy’s own rollouts on 20,000 evaluation paths. Our log recorded the gate as failed on the first trained policy (MAD 0.074 under the first design). The same network passes under either other design, as do all six replicates; under the first design one of six passes.

The check the gate lacked. The disagreement has a cause that the gate could have tested directly. At zero cost the optimal policy is Markov in (t, S) , so its output cannot depend on the previous position. Moving only that input from 0 to 1 at each grid point moves the trained networks’ outputs by 0.052 ± 0.019 on average, and by up to 0.21: the networks use an input the target ignores. This shortcut is nearly invisible on the states the policy visits, where the previous position is close to $\Phi(d_1)$, and exposed off them; a grid with the previous position at 0 is realistic only at t_0 . A well-posed gate states its conditioning and its weighting, reports both, and includes an invariance check on every input the target is known not to depend on. We claim this rung in neither direction; it will be re-run under a gate re-specified in this way.

8 Recommendations: executable checks

Each recommendation is an executable check or a reporting rule. The checks are implemented in the repository and pinned by tests, except the re-specified gate of item 5, whose test is written but not yet run.

1. **Replicate streams.** Derive seeds by hashing (stream label, index), pass list seeds, or split one generator; never pass replicate indices as integers to `from_seed`. Check the cross-replicate dispersion of an estimator against its analytic SE; lag-0 correlations and same-seed reruns cannot see shifted streams.
2. **The classical baseline.** Implement the band as a clip to its edges, so that trade-to-centre is unrepresentable, and check that it beats delta hedging on both CVaR-95 and turnover.
3. **Frictions.** Vary volatility error, adverse and favourable separately, as well as cost, each calibrated to the instrument. Check simulated mean effects against closed forms, and report which friction dominates under the risk measure the claim uses.
4. **Replicates.** Size path counts by the tail metric, and seed counts by power from a noise floor measured at the training budget used, carrying the uncertainty of its s.d. Flag failed runs from their loss histories by a criterion fixed in advance; report them and never drop them. Report comparisons below 80% power as inconclusive rather than null.
5. **Acceptance gates.** State the conditioning of every input and the weighting over states, report both natural designs, and test invariance to inputs the target ignores.
6. **Provenance.** Regenerate every number from committed code and pin each claim with a test, including negative tests that keep withdrawn claims from returning.

9 Related work

Deep hedging and its evaluation. Deep hedging optimises a convex risk measure of simulated P&L directly [Buehler et al., 2019]. Recent work studies robustness to distributional shift through adversarial training [He et al., 2025], training across parameter-uncertainty sets [Lütkebohmert et al., 2022], architectures with a built-in no-transaction band [Imaki et al., 2023], valuation adjustments for execution under frictions [Sakuma, 2026], and comparison against cost-aware classical rules on market data [Kumar, 2026]. Reviews note that neural-hedging conclusions depend on the benchmark chosen and on leakage-free data splits [Ruf and Wang, 2020], and that the field lacks a standard benchmark [Pickard and Lawryshyn, 2023]. Our measurements concern the evaluation apparatus rather than any hedging method; they are complementary to these studies, and we make no claim about their results.

Evaluation statistics. Seed variability alone can produce statistically different results from identical configurations [Henderson et al., 2018]; power analysis should set seed counts [Colas et al., 2018]; interval estimates should replace point estimates when runs are few [Agarwal et al., 2021]; and correlated replicates inflate the variance of benchmark estimates [Bouthillier et al., 2021]. Inference from non-independent replicates is a classical error of experimental design [Hurlbert, 1984]. We measure these quantities for a tail-risk objective in hedging, and show a route by which replicates become non-independent with no deliberate choice at all.

Random streams. Counter-based generators give independent streams through distinct keys or disjoint counter ranges [Salmon et al., 2011]; related seeds producing correlated streams is a known defect class [Matsumoto et al., 2007]; and TensorFlow’s guide warns about overlapping streams from `from_seed` [TensorFlow Authors, 2024]. We document the deterministic integer path behind that warning, and its statistical consequence.

Evaluation pitfalls in financial machine learning. At NeurIPS, evaluation validity in finance has centred on look-ahead leakage in backtests of language-model agents [Li et al., 2025]. Ours is the complementary, simulation-side question: when replicates, ground truth and closed forms are all available, what still goes wrong?

10 Limitations

- **One world, one claim type.** European calls under geometric Brownian motion, with no stochastic volatility or jumps; the sweep covers 12 contracts, the other findings one ATM

contract. Market data enter only to calibrate the two friction axes, for the S&P 500 index; VIX is not the implied volatility of a specific strike, and one-year horizons are not calibrated.

- **A framework-specific mechanism.** Pitfall 1 concerns `from_seed` with TensorFlow 2.21’s Philox generator. We have not examined other versions or frameworks; tests pin the mechanism so that a change fails loudly.
- **One noise-floor configuration.** One architecture and one objective (entropic, zero cost) at two budgets, with a constant learning rate; one failure in 30 runs estimates the failure rate only loosely. The noise floor of a cost-aware policy may differ, and seed counts should be re-derived for each comparison from its own measured floor.
- **No learned-policy accuracy is claimed.** The rung-4 gate is being re-specified (§7).
- **No learned-versus-classical comparison is claimed:** the paper concerns the measurement apparatus, not whether deep hedging beats, ties or loses to the band.
- **Reproducibility gaps.** Findings are parameterised in code (`experiments/findings.py`) rather than by the YAML configurations the project specifies, whose runner is not yet implemented. Market data cannot be redistributed, so the calibration regenerates only after a download; its arithmetic is unit-tested on synthetic data.
- **Literature coverage.** Claims about other work are restricted to sources read in full; we make no claim about the practices of published work in general.

Acknowledgments and Disclosure of Funding

[AUTHOR — approve or edit this disclosure.] Substantial portions of the code, the analysis and the text were produced with the assistance of an AI coding assistant (Claude, Anthropic), under the author’s direction. All reported numbers are regenerated by committed code; all except the market calibration are pinned by automated tests; and every claim about other work is recorded, with page references to the source text, in `docs/07-literature-audit.md`. [AUTHOR — funding and competing-interests statement.]

References

- Rishabh Agarwal, Max Schwarzer, Pablo Samuel Castro, Aaron Courville, and Marc G. Bellemare. Deep reinforcement learning at the edge of the statistical precipice. In *Advances in Neural Information Processing Systems 34 (NeurIPS 2021)*, 2021.
- Xavier Bouthillier, Pierre Delaunay, Mirko Bronzi, Assya Trofimov, Brennan Nichyporuk, Justin Szeto, Naz Sepah, Edward Raff, Kanika Madan, Vikram Voleti, Samira Ebrahimi Kahou, Vincent Michalski, Dmitriy Serdyuk, Tal Arbel, Chris Pal, Gaël Varoquaux, and Pascal Vincent. Accounting for variance in machine learning benchmarks. In *Proceedings of Machine Learning and Systems 3 (MLSys 2021)*, 2021.
- Hans Buehler, Lukas Gonon, Josef Teichmann, and Ben Wood. Deep hedging. *Quantitative Finance*, 19(8): 1271–1291, 2019. doi: 10.1080/14697688.2019.1571683.
- Zeting Chen, Xinyu Cai, Molei Qin, and Bo An. OPHR: Mastering volatility trading with multi-agent deep reinforcement learning. In *Advances in Neural Information Processing Systems 38 (NeurIPS 2025)*, 2025.
- Cédric Colas, Olivier Sigaud, and Pierre-Yves Oudeyer. How many random seeds? Statistical power analysis in deep reinforcement learning experiments, 2018. arXiv:1806.08295.
- Nicole El Karoui, Monique Jeanblanc-Picqué, and Steven E. Shreve. Robustness of the Black and Scholes formula. *Mathematical Finance*, 8(2):93–126, 1998. doi: 10.1111/1467-9965.00047.
- Guangyi He, Tobias Sutter, and Lukas Gonon. Distributional adversarial attacks and training in deep hedging. In *Advances in Neural Information Processing Systems 38 (NeurIPS 2025)*, 2025. arXiv:2508.14757.
- Peter Henderson, Riashat Islam, Philip Bachman, Joelle Pineau, Doina Precup, and David Meger. Deep reinforcement learning that matters. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 32, 2018. doi: 10.1609/aaai.v32i1.11694.
- Stuart H. Hurlbert. Pseudoreplication and the design of ecological field experiments. *Ecological Monographs*, 54(2):187–211, 1984. doi: 10.2307/1942661.

- Shota Imaki, Kentaro Imajo, Katsuya Ito, Kentaro Minami, and Kei Nakagawa. No-transaction band network: A neural network architecture for efficient deep hedging. *The Journal of Financial Data Science*, 5(2):84–99, 2023. doi: 10.3905/jfds.2023.1.125.
- Sheryan Kumar. Deep hedging under realistic market frictions: A regime-conditional empirical study of dynamic option hedging on bitcoin options, 2026. arXiv:2608.29025.
- Hayne E. Leland. Option pricing and replication with transactions costs. *The Journal of Finance*, 40(5):1283–1301, 1985. doi: 10.1111/j.1540-6261.1985.tb02383.x.
- Changlun Li, Yao Shi, Chen Wang, Qiqi Duan, Runke Ruan, Weijie Huang, Haonan Long, Lijun Huang, Nan Tang, and Yuyu Luo. Time travel is cheating: Going live with DeepFund for real-time fund investment benchmarking. In *Advances in Neural Information Processing Systems 38 (NeurIPS 2025), Datasets and Benchmarks Track*, 2025.
- Eva Lütkebohmert, Thorsten Schmidt, and Julian Sester. Robust deep hedging. *Quantitative Finance*, 22(8): 1465–1480, 2022. doi: 10.1080/14697688.2022.2056073.
- Makoto Matsumoto, Isaku Wada, Ai Kuramoto, and Hyo Ashihara. Common defects in initialization of pseudorandom number generators. *ACM Transactions on Modeling and Computer Simulation*, 17(4), 2007. doi: 10.1145/1276927.1276928.
- Reilly Pickard and Yuri Lawryshyn. Deep reinforcement learning for dynamic stock option hedging: A review. *Mathematics*, 11(24):4943, 2023. doi: 10.3390/math11244943.
- R. Tyrrell Rockafellar and Stanislav Uryasev. Optimization of conditional value-at-risk. *Journal of Risk*, 2(3): 21–41, 2000.
- Johannes Ruf and Weiguan Wang. Neural networks for option pricing and hedging: a literature review. *Journal of Computational Finance*, 2020. doi: 10.21314/JCF.2020.390.
- Takayuki Sakuma. Robust hedging valuation adjustment for deep hedging policies under market frictions, 2026. arXiv:2607.25258.
- John K. Salmon, Mark A. Moraes, Ron O. Dror, and David E. Shaw. Parallel random numbers: as easy as 1, 2, 3. In *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis (SC '11)*, pages 1–12, 2011. doi: 10.1145/2063384.2063405.
- TensorFlow Authors. Random number generation. TensorFlow Core guide, https://www.tensorflow.org/guide/random_numbers, 2024. Last updated 2024-08-15; accessed 2026-09-26.
- TensorFlow Authors. `tensorflow/python/ops/stateful_random_ops.py`. TensorFlow 2.21.0 source code, 2026. Function `_make_1d_state` and docstring of `Generator.from_seed`; accessed 2026-09-26.
- A. Elizabeth Whalley and Paul Wilmott. An asymptotic analysis of an optimal hedging model for option pricing with transaction costs. *Mathematical Finance*, 7(3):307–324, 1997. doi: 10.1111/1467-9965.00034. Page references are to the working-paper version revised 7 March 1997.
- Elizabeth Whalley and Paul Wilmott. The best hedging strategy. Mathematical Finance paper 1999mf17, Oxford Financial Research Centre, 1999.

A Reproducibility

Environment. TensorFlow 2.21, Keras 3.15, NumPy 2.2.6, SciPy, Python 3.12, on a laptop CPU; no GPU was used. Code: <https://github.com/Adi7710/deep-hedging-benchmark>. [AUTHOR — record the commit hash and the test-suite result at submission.]

Commands and tests. Commands run from the repository root; adding `--all --json <path>` to `python -m experiments.findings` runs every finding and writes a machine-readable record. Tests (under `tests/`) pin each claim’s direction, ordering and significance at reduced sample sizes, not its exact value.

Tables 1 and 2 (seeding). Command: `python -m experiments.findings seeding`. Tests (`test_seeding.py`): `test_integer_seed_is_written_into_the_philox_counter`, `test_consecutive_integer_seeds_are_shifted_copies`, `test_lag_zero_correlation_cannot_see_the_overlap`, `test_a_list_seed_lands_in_the_key`, `test_hashed_replicates_are_not_shifted_copies`, `test_cross_seed_dispersion_matches_analytic_standard_error`.

Table 3 (band rebalancing). Command: `python -m experiments.findings baseline`. Tests (`test_findings.py`): `test_band_edge_beats_band_centre_beats_delta`, `test_the_edge_rule_advantage_is_significant_and_consistent`, `test_the_centre_over_edge_ratio_is_not_estimable`; and (`test_rungs_3_to_6.py`) `test_band_hedging_beats_naive_delta_under_costs`.

Table 4, Figure 1, Tables 7–8 (friction sweep). Command: `python -m experiments.findings misspecification_sweep`, stored in `paper/findings_sweep.json`; figure by `python paper/workshop/make_figure.py`. Tests: `test_adverse_vol_error_dominates_index_like_costs_in_every_contract`, `test_favourable_vol_error_is_a_gain`, `test_the_ordering_flips_at_illiquid_underlying_costs`, `test_daily_rebalancing_narrows_the_ratio`.

Table 9 (friction mechanism). Command: `python -m experiments.findings friction_mechanism`. Tests: `test_mean_effects_match_closed_forms`, `test_the_tail_metric_amplifies_vol_error_more_than_cost`, `test_which_friction_dominates_depends_on_the_risk_measure`.

Table 10 (market calibration). Command: `python -m experiments.market_calibration --cache-dir <outside the repo> --json paper/market_calibration.json` (downloads public data; raw files never enter the repository). Tests (`test_market_calibration.py`) pin the realised-volatility windows, the cost conversion, bootstrap determinism, and the refusal to cache raw data in a tracked path.

Section 6 (evaluation noise). Commands: `python -m experiments.findings precision_by_n` (100 replicates) and `python -m experiments.findings precision` (paired s.d. and pairing factors). Tests: `test_tail_metric_se_exceeds_the_mean_se_several_fold`, `test_cvar_is_substantially_noisier_than_the_mean`, `test_common_random_numbers_help_but_only_modestly`.

Table 5 and the 8,000-step result (noise floor and power). Commands: `python -m experiments.findings noise_floor` (30 replicates, 2,000 steps) and `noise_floor_long` (8,000 steps; both report each run’s loss-history diagnostics and the robust spread). Tests: `test_ci_halfwidth_uses_the_t_critical_value`, `test_the_ci_halfwidth_is_not_an_80pct_power_mde`, `test_power_reproduces_an_independent_calculation`, `test_learned_vs_learned_needs_more_seeds_than_vs_a_fixed_comparator`, `test_the_loss_history_check_catches_a_run_that_regressed`, `test_robust_z_flags_a_planted_failed_run`.

Table 6 (acceptance gate). Command: `python -m experiments.findings evaluation_design`.

Section 2 checks. `test_mc_price_matches_black_scholes` (`test_rung1_mc_price.py`); `python -m experiments.findings convergence` and `test_delta_hedge_error_shrinks_with_rebalancing` (`test_rung2_pnl_accounting.py`); `test_transaction_costs_includes_final_liquidation`; `test_empirical_cvar_matches_the_training_objective_at_its_optimum` (`test_metrics.py`); `test_band_width_scales_as_cube_root_of_cost`.

Compute. Every experiment ran on a laptop CPU. Compiling the training step with `tf.function` cut the time per gradient step from 92.4 ms to 5.5 ms (16.9×), so a 2,000-step training run takes about 11 s; compiled and eager training give the same loss history to within 10^{-4} (`test_compiled_and_eager_training_agree`). The largest findings (the 30-replicate noise floor, the friction sweep) take minutes to tens of minutes each.

B Full friction sweep

Tables 7 and 8 give every effect of §5 as mean (s.d. across 3 seeds, in units of 10^{-3}). Each effect is a paired difference in CVaR-95 (P&L orientation) against the correctly specified zero-cost delta hedge of the same contract; 40,000 paths per seed.

Table 7: Volatility-error effects: realised $\sigma_r = \rho \sigma_h$, $\sigma_h = 0.2$, zero cost. Mean (s.d. $\times 10^3$).

K	T		n	$\rho = 0.8$	$\rho = 0.9$	$\rho = 1.1$	$\rho = 1.25$
90	0.25	W	13	+1.159 (12.03)	+0.634 (3.54)	-0.687 (3.99)	-1.792 (11.65)
90	1	W	52	+1.871 (7.64)	+1.086 (5.54)	-1.232 (9.67)	-3.241 (24.40)
100	0.25	W	13	+1.410 (8.06)	+0.743 (3.08)	-0.799 (2.60)	-2.073 (6.41)
100	1	W	52	+2.175 (19.45)	+1.222 (9.82)	-1.370 (7.70)	-3.598 (13.24)
110	0.25	W	13	+1.415 (4.90)	+0.768 (0.51)	-0.829 (5.64)	-2.164 (16.50)
110	1	W	52	+2.208 (14.71)	+1.296 (7.42)	-1.471 (7.38)	-3.872 (10.86)
90	0.25	D	63	+0.714 (7.09)	+0.466 (3.89)	-0.559 (6.77)	-1.496 (11.38)
90	1	D	252	+1.183 (14.19)	+0.849 (11.79)	-1.136 (4.89)	-3.032 (20.67)
100	0.25	D	63	+1.048 (5.61)	+0.601 (5.14)	-0.681 (4.72)	-1.790 (8.41)
100	1	D	252	+1.493 (13.68)	+0.984 (10.31)	-1.276 (6.53)	-3.397 (18.95)
110	0.25	D	63	+0.879 (6.93)	+0.572 (5.76)	-0.681 (12.95)	-1.818 (21.17)
110	1	D	252	+1.336 (0.84)	+0.989 (0.93)	-1.359 (5.93)	-3.637 (8.02)

Table 8: Cost effects: proportional cost c , $\sigma_r = \sigma_h = 0.2$. Mean (s.d. $\times 10^3$).

K	T		n	1 bp	5 bp	25 bp	50 bp
90	0.25	W	13	-0.026 (0.11)	-0.131 (0.50)	-0.664 (1.87)	-1.347 (3.65)
90	1	W	52	-0.041 (0.08)	-0.206 (0.30)	-1.052 (1.20)	-2.142 (4.58)
100	0.25	W	13	-0.027 (0.18)	-0.133 (0.97)	-0.679 (3.76)	-1.386 (6.58)
100	1	W	52	-0.043 (0.34)	-0.218 (1.69)	-1.114 (5.71)	-2.274 (9.28)
110	0.25	W	13	-0.023 (0.14)	-0.117 (0.69)	-0.598 (3.88)	-1.227 (8.08)
110	1	W	52	-0.044 (0.23)	-0.221 (1.08)	-1.134 (5.60)	-2.320 (11.16)
90	0.25	D	63	-0.042 (0.24)	-0.210 (0.97)	-1.087 (2.90)	-2.234 (3.63)
90	1	D	252	-0.075 (0.36)	-0.382 (2.00)	-2.003 (8.72)	-4.121 (12.40)
100	0.25	D	63	-0.046 (0.18)	-0.235 (0.59)	-1.218 (3.24)	-2.503 (5.85)
100	1	D	252	-0.082 (0.43)	-0.416 (1.61)	-2.185 (6.00)	-4.498 (14.06)
110	0.25	D	63	-0.041 (0.11)	-0.210 (0.59)	-1.110 (1.54)	-2.302 (1.59)
110	1	D	252	-0.084 (0.25)	-0.429 (1.45)	-2.265 (4.33)	-4.676 (7.16)

C The friction ordering, decomposed

Leland’s adjustment prices a one-way proportional cost c , paid at rebalances Δt apart, as extra variance $\sigma^2(1 + \text{Le})$, $\text{Le} = c\sqrt{8/\pi}/(\sigma\sqrt{\Delta t})$ [Leland, 1985]. With $\mu = r = 0$ and $G = \mathbb{E}[\frac{1}{2} \int_0^T \Gamma S^2 \sigma^2 dt] = \sigma \text{Vega}/2$, the mean effects have closed forms: the volatility error costs exactly $V(\sigma_h) - V(\sigma_r)$, and the cost is, to first order, $\text{Le} G + c S_0 (\Delta_0 + \Phi(d_1))$ — rebalancing plus opening and closing the hedge, since $\mathbb{E}[S_n \delta_{n-1}] \approx S_0 \Phi(d_1)$ when $\mu = 0$. Table 9 compares both with simulation, on the draws of Table 4, and sets the mean ratios beside the CVaR-95 ratios.

D Market calibration

Realised volatility is $\text{RV}_t = \sqrt{(252/21) \sum_{k=1}^{21} r_{t+k}^2}$ from S&P 500 closes (not demeaned), set against VIX at the close of day t ; the ratio is exactly the sweep’s ρ when the hedge runs at implied volatility. Overlapping windows are handled by a circular block bootstrap (252-day blocks; 63 and 504 give the same interval to 0.5 percentage points) and by the 21 non-overlapping subsamples. A proportional one-way cost for an instrument quoted one tick wide is $c = (\text{tick}/2)/\text{price}$. Sources and licensing are recorded in `experiments/market_calibration.py`; the headline was recomputed independently from Cboe’s own VIX and SPX files.

Table 9: Mean versus CVaR-95 effects, 3 seeds of 40,000 paths. Vol: effect of $\rho = 1.1$. 5 bp: effect of cost. Sim/exact and sim/pred: simulated mean against the closed form. Rebal.: share of the 5 bp cost spent on rebalancing (the rest opens and closes the hedge). Amp.: CVaR-95 effect over mean effect. Ratios: $|vol|$ over $|cost|$ at 5 and 25 bp.

K	T		n	Vol mean		5 bp mean		Amp.			Ratio, 5 bp		Ratio, 25 bp	
				sim	exact	sim	pred	Rebal.	vol	5 bp	mean	CVaR-95	mean	CVaR-95
90	0.25	W	13	-0.228	-0.228	-0.1161	-0.1177	25%	3.01	1.13	1.97	5.24	0.39	1.03
90	1	W	52	-0.663	-0.664	-0.1667	-0.1678	56%	1.86	1.24	3.98	5.97	0.80	1.17
100	0.25	W	13	-0.399	-0.398	-0.1066	-0.1093	51%	2.00	1.25	3.74	6.00	0.75	1.18
100	1	W	52	-0.792	-0.793	-0.1669	-0.1682	68%	1.73	1.31	4.74	6.29	0.95	1.23
110	0.25	W	13	-0.277	-0.277	-0.0545	-0.0565	66%	2.99	2.14	5.08	7.11	1.02	1.39
110	1	W	52	-0.751	-0.751	-0.1413	-0.1422	75%	1.96	1.56	5.31	6.65	1.06	1.30
90	0.25	D	63	-0.228	-0.228	-0.1546	-0.1552	44%	2.45	1.36	1.48	2.66	0.30	0.51
90	1	D	252	-0.663	-0.664	-0.2806	-0.2811	74%	1.71	1.36	2.36	2.98	0.47	0.57
100	0.25	D	63	-0.397	-0.398	-0.1770	-0.1782	71%	1.71	1.33	2.25	2.90	0.45	0.56
100	1	D	252	-0.792	-0.793	-0.3047	-0.3054	82%	1.61	1.37	2.60	3.07	0.52	0.58
110	0.25	D	63	-0.277	-0.277	-0.1018	-0.1023	82%	2.46	2.07	2.72	3.23	0.54	0.61
110	1	D	252	-0.751	-0.751	-0.2704	-0.2707	87%	1.81	1.59	2.78	3.17	0.56	0.60

Table 10: Calibration of the two friction axes (strike dates 1990-01-02 to 2026-08-26).

Quantity	Value
Windows (overlapping; approx. independent)	9,227; 440
Median RV/VIX	0.73
$\Pr(RV < 0.9 \text{ VIX})$, favourable	78.1%
$\Pr(RV > 1.1 \text{ VIX})$ [95% block-bootstrap CI]	8.8% [6.3, 11.4]
$\Pr(RV > 1.1 \text{ VIX})$, range over non-overlapping offsets	7.5–10.0%
$\Pr(RV > 1.25 \text{ VIX})$	4.9%
One-tick half-spread, E-mini S&P 500 (2026-09-25; 1998–2026 range)	0.16 bp (0.16–1.85)
One-cent half-spread, SPY (2026-09-25)	0.065 bp
Quoted half-spread, median S&P 500 constituent (Dec. 2015)	1.6 bp
Quoted half-spread, \$100M–\$1bn stocks (2013 medians)	7.8–61 bp